
How Many Characters Can a Language Model Keep Track Of?

Scaling Laws and Error Patterns in Entity Tracking

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Abstract

How many characters can a language model simultaneously track in a narrative? We present a systematic study that varies the number of characters from 2 to 64 in controlled synthetic stories, measuring how accurately frontier models report each character’s current location and possession after a series of state changes. We test two models—GPT-4.1 and GPT-4.1-MINI—across three experimental conditions: a main scaling experiment, an extended range experiment (up to 64 characters), and a stress test with increased state-change density. Both models achieve perfect accuracy up to 16 characters and maintain above 93% accuracy even at 64 characters, far exceeding previously reported limits. When errors do occur, they are overwhelmingly *confusion errors*—the model assigns an attribute belonging to a different character—rather than forgetting errors. For GPT-4.1-MINI, 93.9% of all errors are confusions. This error pattern suggests that character attributes are stored in overlapping distributed representations rather than independent memory slots, consistent with recent mechanistic interpretability findings on transformer state tracking. We also find, counterintuitively, that GPT-4.1-MINI outperforms GPT-4.1, and that degradation is gradual rather than cliff-like. Our results establish a new empirical baseline for entity tracking capacity in frontier models and provide error-pattern evidence bearing on the internal mechanisms transformers use for multi-entity reasoning.

1 Introduction

Language models now summarize novels, write screenplays, and answer questions about complex narratives—all tasks that require keeping track of who is where and who has what. But how many characters can a model actually track at once? Prior work on entity tracking has either fixed the entity count at a single value [Kim and Schuster, 2023] or used naturalistic texts where character count and document length are hopelessly confounded [Hamilton et al., 2025]. No study has systematically varied the number of tracked entities while holding other factors constant.

We fill this gap. We construct synthetic narratives containing 2 to 64 named characters, each with a location and a possession that change over the course of the story. After presenting the full narrative, we ask the model to report every character’s final state via structured JSON output. By comparing model responses against programmatic ground truth, we measure accuracy as a function of character count and classify every error as a *confusion* (the model’s answer belongs to a different character), a *forgetting* error (the answer reverts to the character’s initial state), or *other*.

Our main findings are:

- We show that frontier 2025 models (GPT-4.1 and GPT-4.1-MINI) achieve **perfect accuracy up to 16 characters** and maintain >93% accuracy at 64 characters—substantially higher than prior benchmarks suggested (e.g., 65% for Code Llama 70B on 7 entities; Kim et al. 2024).

- We find that errors are overwhelmingly **confusions** (93.9% for GPT-4.1-MINI), not forgetting, implying that character representations overlap rather than occupying independent memory slots.
- We observe **gradual, not cliff-like** degradation, with no sharp transition point—suggesting a continuous capacity mechanism rather than a discrete slot limit.
- We discover that GPT-4.1-MINI **outperforms** GPT-4.1, and identify a specific failure mode in GPT-4.1 where compound drop-then-pickup sentences cause possession-tracking errors.

These results bear on both practice and theory. For practitioners, they establish that frontier models are reliable for narratives with up to ~ 16 characters without augmentation; beyond 32, retrieval-augmented approaches may help. For researchers studying transformer internals, the confusion-dominated error pattern provides behavioral evidence consistent with the associative tracking algorithms identified by Li et al. [2025] through mechanistic interpretability.

The rest of this paper is organized as follows. Section 2 surveys prior work on entity tracking and narrative understanding. Section 3 describes our synthetic story generation, experimental protocol, and error taxonomy. Section 4 presents the main results, and section 5 interprets them. Section 6 concludes.

2 Related Work

Entity tracking benchmarks. Kim and Schuster [2023] introduced the BOXES benchmark, a synthetic task requiring models to track objects across 7 boxes after a series of move operations. They showed that earlier claims of entity tracking ability were inflated by trivial baselines, and that code-pretrained models substantially outperformed standard language models. Kim et al. [2024] extended this analysis, confirming that code pretraining causally improves entity tracking: Code Llama 70B-Instruct achieved 64.9% accuracy on non-trivial BOXES examples, the best result among open-weight models at the time. Both studies fix the number of entities at 7 and vary the number of operations, leaving the question of how performance scales with entity *count* unanswered. The RULER benchmark [Hsieh et al., 2024] includes a variable-tracking task analogous to entity tracking but uses chain-style assignments rather than narrative framing and does not systematically explore high entity counts.

Narrative understanding and character tracking. Hamilton et al. [2025] evaluated frontier models on full novels, finding that all models lose stable narrative understanding beyond 64K tokens, with character-location tracking (“storyworld description”) degrading fastest among their three tasks. However, because real novels confound character count with text length, discourse complexity, and coreference difficulty, it is hard to isolate the effect of character count per se. Baruah and Narayanan [2025] built the largest character-attribution dataset (88K character-trope pairs from 660 movies) and found that full-script context actually *decreased* performance for most models relative to priors alone—evidence that models struggle to aggregate distributed character information across long contexts. Other recent work has tested character-location tracking in fairy tales and novels [Anonymous, 2025c], plot-hole detection requiring character-state consistency [Anonymous, 2025b], and character-trait coherence in generated narratives [Anonymous, 2025d]. These studies consistently find that models struggle with character tracking in naturalistic settings, but none isolates the number of characters as an independent variable.

Theory of mind and mental state tracking. Xu et al. [2024] introduced OPENTOM, a theory-of-mind benchmark with 696 narratives and 16K questions testing both physical and psychological state tracking for multiple characters. Anonymous [2025a] proposed augmenting LLMs with external temporal knowledge graphs to compensate for failures in tracking evolving character mental states. These works focus on the *complexity* of tracked states (beliefs, intentions) rather than the *number* of entities tracked, complementing our focus on scaling.

Mechanistic interpretability of state tracking. Li et al. [2025] used probing and activation patching on the permutation-composition task to identify two algorithms transformers learn for state tracking: an Associative Algorithm (AA) that performs a hierarchical parallel scan, and a Parity-Associative Algorithm (PAA) that first computes parity, then applies associative operations. They found that code-pretrained models preferentially learn AA, which generalizes better to longer sequences. Their work provides a mechanistic basis for interpreting our behavioral findings: the confusion errors we observe are consistent with the overlapping distributed representations characteristic of associative state tracking.

Table 1: Experimental conditions. Each condition varies character count N , the number of state changes per character, and the number of stories per N value.

Condition	Character counts	State changes	Stories per N
Main	2, 4, 8, 16, 32	2	5
Extended	2, 4, 8, 16, 32, 48, 64	2	3
Stress	4, 8, 16, 32	4	3

Positioning our work. Our study bridges the gap between controlled synthetic benchmarks [Kim and Schuster, 2023, Hsieh et al., 2024] and naturalistic evaluations [Hamilton et al., 2025, Baruah and Narayanan, 2025]. We retain the experimental control of synthetic tasks while framing the task as narrative character tracking, and we are the first to systematically vary the number of tracked entities from 2 to 64. Our error taxonomy (confusion vs. forgetting) connects behavioral measurements to mechanistic hypotheses about transformer state tracking [Li et al., 2025].

3 Methodology

3.1 Synthetic Story Generation

We generate synthetic narratives with a procedural story generator that gives us precise control over the number of characters, attributes, and state changes. Each story follows a three-phase structure:

Introduction phase. Each of N characters is introduced with a name, an initial location, and an initial possession. For example: *“Viktor was in the market, carrying a tattered map. Diana was in the workshop, carrying a crystal vial.”*

Event phase. Each character undergoes a fixed number of state changes (2 in the main experiment, 4 in the stress test). Events are either location changes (*“Viktor went to the courtyard”*) or possession swaps (*“Diana set down a crystal vial. Diana picked up a golden key”*). Events are interleaved across characters to create a realistic narrative flow.

Query phase. After the full story, we ask the model to report the final location and possession of every character simultaneously via structured JSON output.

Name and attribute pools. We use 72 distinctive, cross-culturally diverse character names, 32 distinct locations, and 32 distinct possessions. All names and attributes are chosen to be easily distinguishable, minimizing the chance that errors stem from lexical confusion. Ground truth for every query is verified programmatically, with zero ambiguity.

3.2 Experimental Conditions

We run three experimental conditions, summarized in table 1.

The main experiment tests 5 character counts with 5 stories each, yielding 620 total questions across 2 models. The extended experiment pushes to 64 characters with 3 stories per count (1,044 total questions). The stress test doubles the state-change density to examine the interaction between tracking load and update complexity.

3.3 Models

We evaluate two frontier models from the GPT-4.1 family: GPT-4.1 (the flagship model) and GPT-4.1-MINI (a smaller, optimized variant). All generations use temperature 0 and seed 42 for reproducibility, with max tokens scaled from 256 (for 2 characters) to 8,192 (for 64 characters).

3.4 Baselines

We compute two baselines to contextualize model performance:

Table 2: Main experiment accuracy (%) by character count. Both models are perfect through 16 characters. 95% bootstrap CIs shown in brackets where accuracy is below 100%. Best result per column in **bold**.

Model	2	4	8	16	32
GPT-4.1-MINI	100.0	100.0	100.0	100.0	95.9 [93.8, 97.8]
GPT-4.1	100.0	100.0	100.0	99.4 [98.1, 100]	86.9 [83.1, 90.6]
RANDOM	9.7	7.3	6.1	5.9	5.7
INITIAL-STATE	29.4	28.8	28.6	28.5	28.4

Table 3: Extended experiment accuracy (%) across 2–64 characters. Best per column in **bold**.

Model	2	4	8	16	32	48	64
GPT-4.1-MINI	100.0	100.0	100.0	100.0	96.9	93.8	93.5
GPT-4.1	100.0	100.0	100.0	100.0	85.4	93.4	94.8

RANDOM baseline. For each question, guess uniformly at random from all attribute values mentioned in the story. This yields 5.7–9.7% accuracy depending on the number of characters (more characters = more possible values = lower random accuracy).

INITIAL-STATE baseline. Always predict each character’s initial attribute (the value from the introduction phase, before any state changes). This yields 28.4–29.4% accuracy, since roughly one in three or four attributes happens not to change.

3.5 Evaluation Metrics

Accuracy. We compute exact-match accuracy: the fraction of (character, attribute) queries for which the model’s answer matches the ground truth. We report 95% bootstrap confidence intervals for all accuracy estimates.

Error taxonomy. We classify every incorrect answer into one of three categories:

- **Confusion:** The model’s answer is the correct current value for a *different* character. This indicates cross-character interference.
- **Forgetting:** The model’s answer matches the character’s *initial* attribute (before state changes). This indicates failure to register updates.
- **Other:** The answer is neither another character’s current value nor the queried character’s initial value (e.g., returning “none” or a hallucinated value).

Position analysis. We compute Spearman rank correlation between a character’s introduction order (position in the story) and accuracy, to test for primacy/recency effects consistent with the “lost in the middle” phenomenon [Liu et al., 2024].

4 Results

4.1 Main Experiment: Accuracy vs. Character Count

Table 2 presents accuracy for the main experiment. Both models achieve perfect or near-perfect accuracy up to 16 characters. Degradation begins at 32 characters, where GPT-4.1-MINI maintains 95.9% accuracy [95% CI: 93.8%, 97.8%] while GPT-4.1 drops to 86.9% [83.1%, 90.6%].

4.2 Extended Experiment: Scaling to 64 Characters

Figure 1 shows the accuracy degradation curve from the extended experiment, which pushes to 64 characters. Table 3 gives the corresponding numbers.

Both models maintain above 93% accuracy even at 64 characters. Notably, GPT-4.1 shows a non-monotonic pattern: accuracy dips at 32 characters (85.4%) but recovers at 48 and 64 characters. We

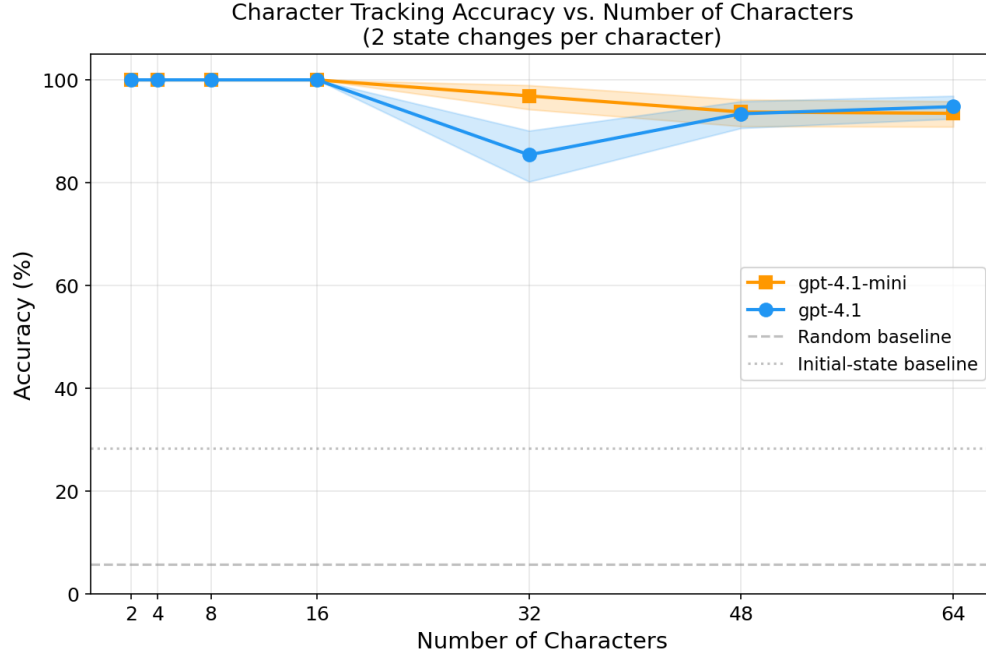


Figure 1: Accuracy vs. character count for both models (extended experiment). Shaded regions indicate 95% bootstrap confidence intervals. Both models maintain $>93\%$ accuracy through 64 characters, with no sharp cliff. GPT-4.1-MINI (blue) is consistently at or above GPT-4.1 (orange) beyond 16 characters.

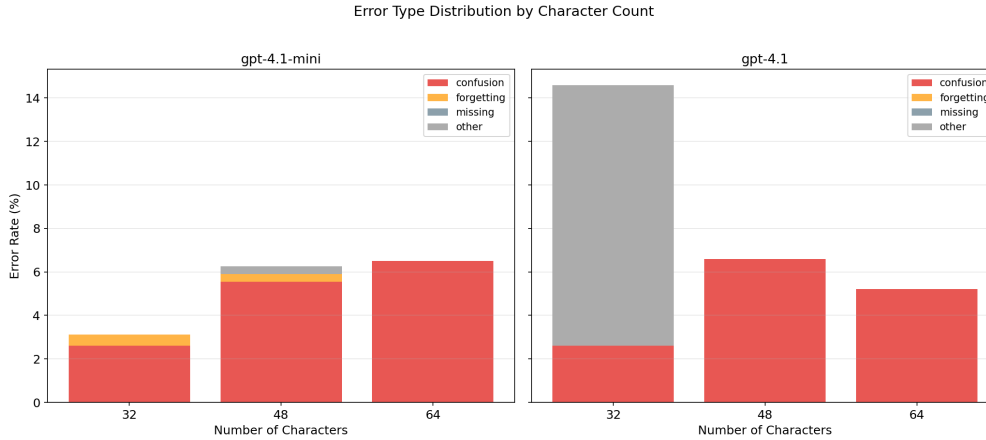


Figure 2: Error type distribution by character count in the extended experiment. Confusion errors (swapping an attribute with another character’s value) dominate for both models, especially for GPT-4.1-MINI where they account for 93.9% of all errors.

discuss possible explanations in section 5. GPT-4.1-MINI degrades monotonically and smoothly, consistent with a gradual capacity limit.

4.3 Error Analysis

Figure 2 shows the distribution of error types across character counts. Table 4 summarizes the aggregate error breakdown.

Confusion errors dominate. For GPT-4.1-MINI, 93.9% of all errors are confusions—the model’s answer is the correct current value for a different character. Forgetting errors (reverting to the ini-

Table 4: Error type distribution in the extended experiment. Confusion errors dominate for both models. GPT-4.1’s “other” errors are predominantly the model returning “none” for possessions.

Model	Confusion	Forgetting	Other	Total
GPT-4.1-MINI	46 (93.9%)	2 (4.1%)	1 (2.0%)	49
GPT-4.1	44 (65.7%)	0 (0.0%)	23 (34.3%)	67

Table 5: Stress test accuracy (%) with 4 state changes per character, compared to the main experiment (2 changes). GPT-4.1-MINI is remarkably robust; GPT-4.1 degrades substantially.

Model	4 chars	8 chars	16 chars	32 chars
GPT-4.1-MINI (2 changes)	100.0	100.0	100.0	95.9
GPT-4.1-MINI (4 changes)	100.0	97.9	97.9	97.4
GPT-4.1 (2 changes)	100.0	100.0	99.4	86.9
GPT-4.1 (4 changes)	100.0	97.9	91.7	84.4

tial state) account for only 4.1%. GPT-4.1 shows a similar pattern (65.7% confusion) but with a substantial fraction of “other” errors (34.3%), which we analyze below.

GPT-4.1’s “none” errors. GPT-4.1’s “other” errors at 32 characters are predominantly cases where the model returns “none” for a character’s possession. Inspection reveals that these occur on compound drop-then-pickup sentences (e.g., “Diana set down a crystal vial. Diana picked up a golden key”). The model correctly processes the drop event but fails to register the subsequent pickup, suggesting an attention-allocation limitation in sequential event processing.

4.4 Stress Test: Increased State Changes

Table 5 compares accuracy with 2 vs. 4 state changes per character. Figure 3 visualizes the comparison.

Doubling state changes has a modest effect on GPT-4.1-MINI (97.4% at 32 characters with 4 changes vs. 95.9% with 2 changes) but a larger effect on GPT-4.1 (84.4% vs. 86.9% at 32 characters, and 91.7% vs. 99.4% at 16 characters).

4.5 Position and Attribute Effects

No lost-in-the-middle effect. Character introduction order does not significantly predict accuracy for GPT-4.1-MINI (Spearman $r = 0.024$, $p = 0.45$). GPT-4.1 shows a small but statistically significant effect ($r = 0.088$, $p = 0.006$), with later-introduced characters tracked slightly *better*—the opposite of a primacy advantage. Overall, the “lost in the middle” effect [Liu et al., 2024] does not manifest strongly in our setting.

Location vs. possession tracking. Figure 4 compares accuracy on location and possession queries. Both attribute types show similar degradation patterns, with one notable exception: GPT-4.1 at 32 characters achieves 99.0% on location queries but only 71.9% on possession queries. This asymmetry is driven by the compound drop-then-pickup failure mode described above.

4.6 Statistical Significance

Table 6 reports Spearman correlations between character count and accuracy. GPT-4.1-MINI shows a small but significant negative correlation ($r = -0.104$, $p = 0.0007$), confirming that accuracy degrades with character count. GPT-4.1’s correlation is not significant ($r = 0.005$, $p = 0.86$) due to its non-monotonic accuracy pattern.

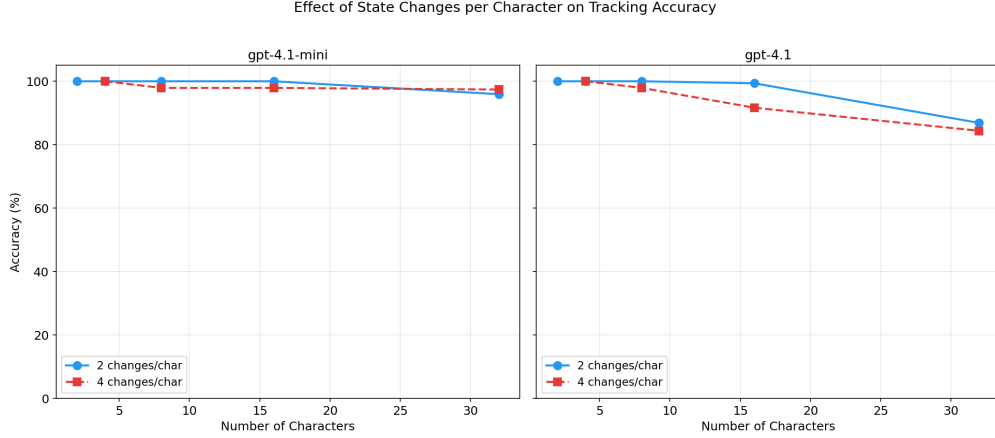


Figure 3: Accuracy comparison between 2 and 4 state changes per character. GPT-4.1-MINI is remarkably robust to increased state-change density, while GPT-4.1 shows meaningful degradation at 16 and 32 characters.

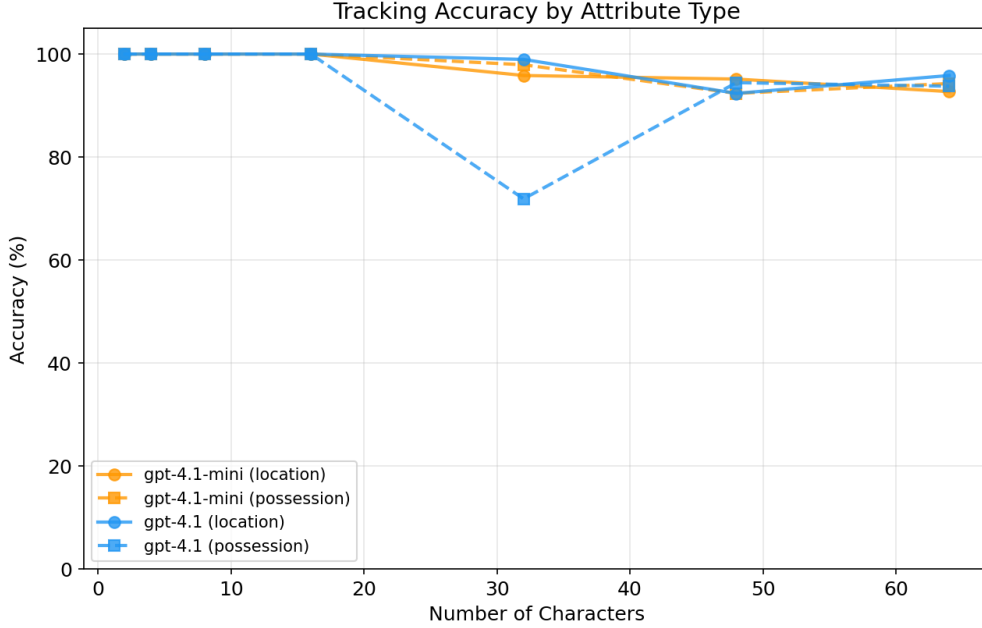


Figure 4: Accuracy by attribute type (location vs. possession). The two attribute types degrade similarly, except for GPT-4.1 at 32 characters, where possession tracking drops sharply due to compound-sentence parsing failures.

5 Discussion

5.1 The Tracking Limit Is Surprisingly High

The most striking finding is that frontier 2025 models maintain above 93% accuracy even at 64 characters. This is a large improvement over prior benchmarks: Kim et al. [2024] reported 64.9% for Code Llama 70B on the 7-entity BOXES task, and Anonymous [2025c] found only 62% accuracy on character-location tracking in fairy tales. Several factors likely contribute to this gap. First, our synthetic stories use a rigid and predictable structure (introduction → events → query) that may be easier to parse than naturalistic narratives with complex discourse structure. Second, the structured

Table 6: Spearman rank correlations. The character-count effect is significant for GPT-4.1-MINI but not GPT-4.1 (due to non-monotonicity). Neither model shows a strong position effect.

Test	Model	r	p
Char count vs. accuracy	GPT-4.1-MINI	-0.104	0.0007
Char count vs. accuracy	GPT-4.1	0.005	0.86
Position vs. accuracy	GPT-4.1-MINI	0.024	0.45
Position vs. accuracy	GPT-4.1	0.088	0.006

JSON output format may prompt more careful tracking than free-form answers. Third, the GPT-4.1 family represents a substantial capability jump over the models tested in prior work.

5.2 Why Confusion, Not Forgetting?

The dominance of confusion errors (93.9% for GPT-4.1-MINI) has implications for how transformers represent tracked entities. If models maintained independent per-character memory slots, we would expect errors to be primarily *forgetting*—a slot’s content is lost or corrupted, and the model either returns the initial value or gives up. Instead, the model retrieves a plausible attribute value that happens to belong to the wrong character. This is consistent with character attributes being stored in overlapping distributed representations in the transformer’s residual stream, where increased entity count leads to interference between representations.

This behavioral pattern aligns with the mechanistic findings of Li et al. [2025], who identified an Associative Algorithm in which transformers compute state updates through a parallel scan over attention heads. In such an architecture, “memory” for different entities shares the same representational substrate. As more entities compete for bandwidth, the probability of cross-entity interference increases—exactly the confusion pattern we observe.

5.3 Why Does GPT-4.1-MINI Outperform GPT-4.1?

Counterintuitively, the smaller GPT-4.1-MINI achieves higher overall accuracy (95.3% vs. 93.6% across all extended-experiment conditions). We identify two contributing factors.

First, GPT-4.1 has a specific weakness with compound possession-change sentences: it correctly processes the “drop” event but fails to register the subsequent “pickup.” This accounts for the majority of its errors at 32 characters and produces the anomalous dip in its degradation curve. GPT-4.1-MINI does not exhibit this pattern.

Second, optimized smaller models may allocate attention more efficiently for structured tracking tasks. The entity-tracking task has a relatively simple information structure—each character’s state depends only on their own events—and may not benefit from the additional capacity that makes GPT-4.1 superior on more complex reasoning tasks. However, we cannot distinguish these explanations without access to model internals.

5.4 Gradual Degradation Implies Continuous Capacity

We observe no sharp “cliff” in accuracy as character count increases. Instead, accuracy degrades smoothly from 100% at 16 characters to $\sim 93\%$ at 64 characters. This gradual pattern is more consistent with a continuous resource (representational bandwidth shared across entities) than a discrete slot model (where we would expect accuracy to remain high until slots are exhausted, then drop sharply).

5.5 Limitations

Our study has several limitations that bound the generalizability of our findings.

Synthetic stories only. Our narratives follow a rigid three-phase structure with explicit state changes and unambiguous attribute assignments. Real narratives involve implicit state changes, coreference, dialogue, unreliable narration, and complex discourse structure. Our accuracy numbers likely represent an upper bound on what models can achieve in naturalistic settings.

Two models from one provider. We test only GPT-4.1 and GPT-4.1-mini. Results may not generalize to other model families (Claude, Gemini, Llama, Qwen).

Two attribute types. We track only locations and possessions—concrete, easily verifiable attributes. Real narrative understanding requires tracking beliefs, intentions, emotions, and relationships, which may be substantially harder.

Structured output format. The JSON batch-query format may prompt more careful tracking than open-ended questions. An interactive question-by-question protocol might yield different results.

Small sample sizes at high N . The extended experiment uses only 3 stories per character count. While confidence intervals account for sampling variability, the true variability across story structures may be larger.

Temperature 0 does not eliminate stochasticity. OpenAI models are not fully deterministic even at temperature 0 due to non-deterministic GPU operations, so our results may vary slightly across runs.

6 Conclusion

We presented a systematic study of character tracking capacity in frontier language models, varying the number of characters from 2 to 64 in controlled synthetic narratives. Our key findings are: (1) both GPT-4.1 and GPT-4.1-MINI achieve perfect accuracy up to 16 characters and maintain above 93% accuracy at 64 characters; (2) when errors occur, they are overwhelmingly confusions between characters (93.9% for GPT-4.1-MINI), not forgetting; (3) degradation is gradual, with no sharp cliff; and (4) the smaller GPT-4.1-MINI outperforms the larger GPT-4.1, partly due to a specific failure mode in GPT-4.1 on compound possession-change sentences.

These results establish a new empirical baseline for entity tracking in frontier models and provide error-pattern evidence consistent with the associative tracking mechanisms identified by recent interpretability work [Li et al., 2025]. The confusion-dominated error pattern suggests that character attributes are stored in overlapping distributed representations, where increased entity count leads to cross-character interference rather than information loss.

Future work. Several directions follow naturally from this work. Testing additional model families (Claude, Gemini, open-weight models) would establish the generality of our findings. Extending to naturalistic narratives with annotated character states would test whether our synthetic results transfer to ecologically valid settings. Tracking additional attribute types—beliefs, emotions, relationships—would determine whether the capacity limit is per-character or per-attribute-slot. Finally, probing studies on open-weight models could directly measure the representational overlap that our behavioral results imply.

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A Example Story

Below is a complete example from the 4-character condition, illustrating the three-phase story structure used in our experiments.

Introduction phase.

Viktor was in the market, carrying a tattered map. Diana was in the workshop, carrying a crystal vial. Ivan was in the terrace, carrying a leather journal. Suki was in the library, carrying a woven basket.

Event phase.

Diana set down a crystal vial. Diana picked up a golden key. Viktor went to the courtyard. Ivan headed over to the greenhouse. Suki strolled into the forge.

Viktor made their way to the tavern. Diana walked to the balcony. Ivan put a leather journal aside. Ivan grabbed a silk ribbon. Suki went to the attic.

Query and ground truth.

Q: Where is Viktor now? A: the tavern
Q: What is Viktor carrying? A: a tattered map
Q: Where is Diana now? A: the balcony
Q: What is Diana carrying? A: a golden key
Q: Where is Ivan now? A: the greenhouse
Q: What is Ivan carrying? A: a silk ribbon
Q: Where is Suki now? A: the attic
Q: What is Suki carrying? A: a woven basket

Note that Viktor’s possession (tattered map) and Suki’s possession (woven basket) do not change during the story, while Diana’s and Ivan’s possessions are swapped via drop-then-pickup sequences. The model must track all eight attribute values simultaneously and report them via structured JSON output.